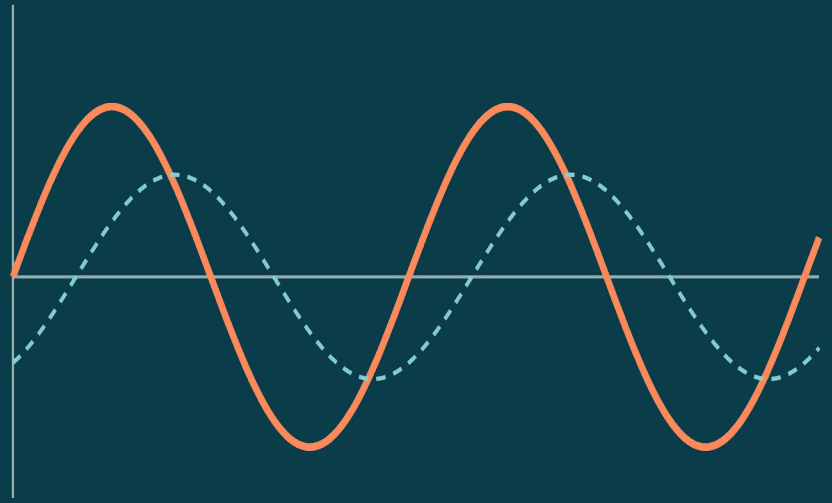
555 Timer and 78XX Regulators



Basic Electrical and Electronics Engineering

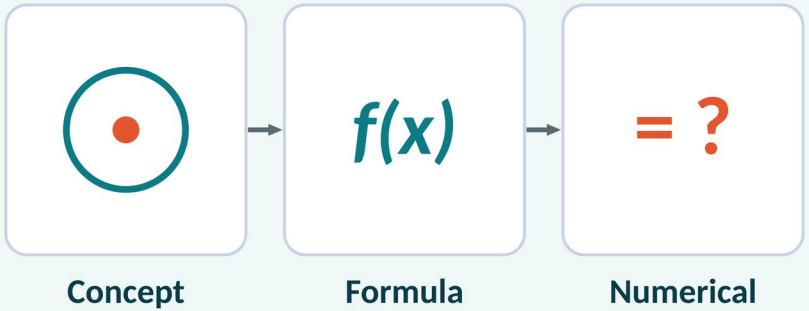
Unit 5: Op-Amp and Linear IC Applications

Topic: 555 timer IC in astable mode; 78XX series IC voltage regulators

Prof. Mitesh Astik

Learning Outcomes

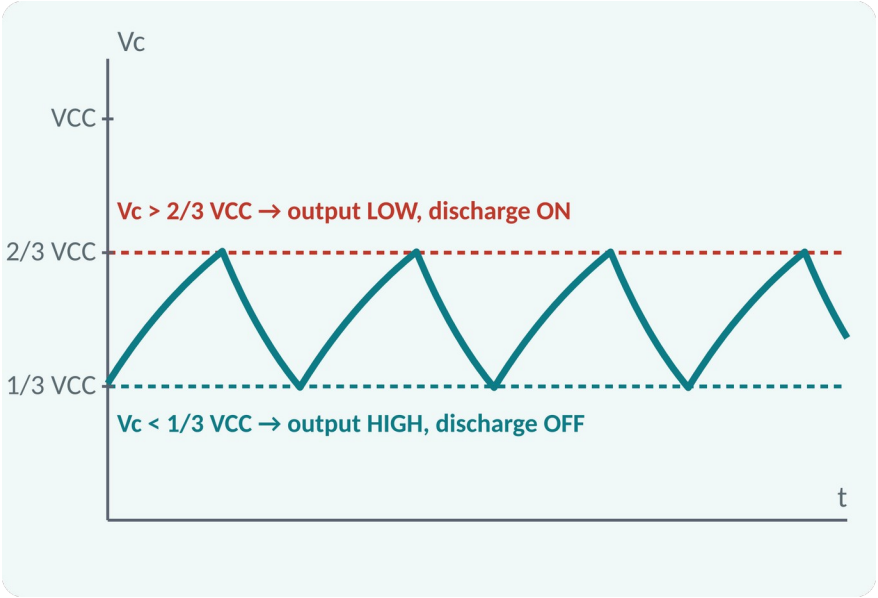
- Describe the internal blocks and pins of the 555.
- Explain astable operation with RA, RB and C.
- Calculate frequency and duty cycle.
- Use 78XX regulators in a power supply.



Linked to CO5
Describe the working principles of op-amp and signal-processing circuits.

Recap: What You Already Know

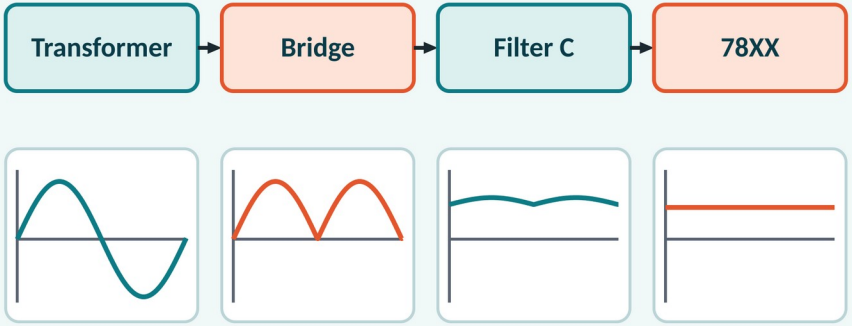
- Capacitor charges as v = V(1 − e^(−t/RC)).
- Time constant τ = RC.
- Comparator output ±Vsat.
- Voltage divider: Vout = Vin·R2/(R1 + R2).



Capacitor charging between two levels takes t = RC·ln[(Vf − V1)/(Vf − V2)]

Motivation: Timers, Flashers and Regulators

- Flashing lights, buzzers, clock pulses.
- Pulse generation for digital circuits.
- Every power supply needs a regulator.
- Both are cheap, popular ICs.



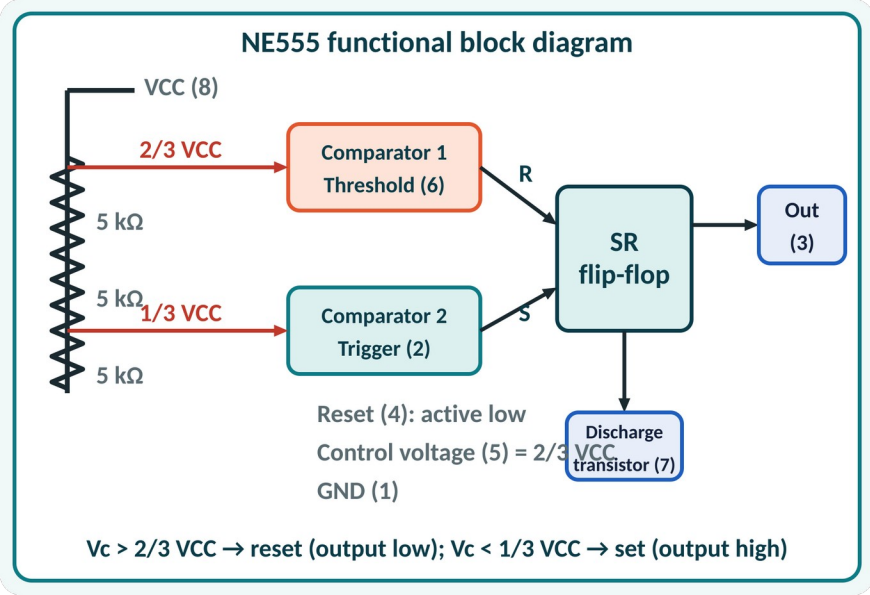
230 V AC → 12–15 V AC → pulsating DC → smoothed DC → fixed 5 V DC

Regulator keeps Vo constant for changes in Vin and load

555: pulse generator; 78XX: steady DC supply

555 Timer: Internal Structure

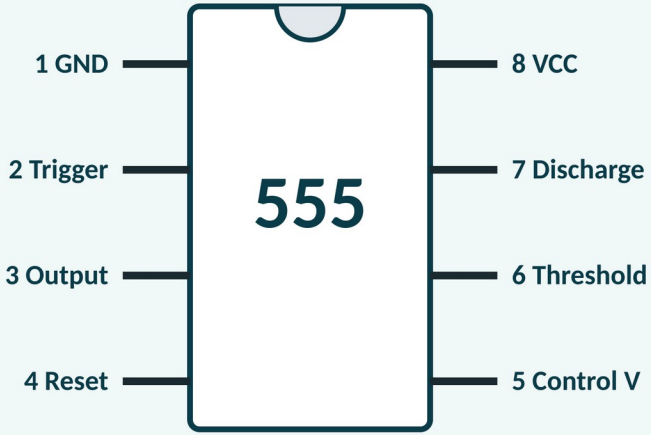
- Voltage divider of three 5 kΩ resistors.
- Comparator 1 (threshold) compares with 2/3 VCC.
- Comparator 2 (trigger) compares with 1/3 VCC.
- SR flip-flop stores the state.
- Output stage and discharge transistor.



Thresholds: upper = 2/3 VCC, lower = 1/3 VCC

555 Pin Configuration

- Pin 1: ground; pin 8: VCC.
- Pin 2: trigger (set at 1/3 VCC).
- Pin 6: threshold (reset at 2/3 VCC).
- Pin 3: output; pin 4: reset (tie to VCC).
- Pin 7: discharge; pin 5: control voltage (0.01 µF to ground).

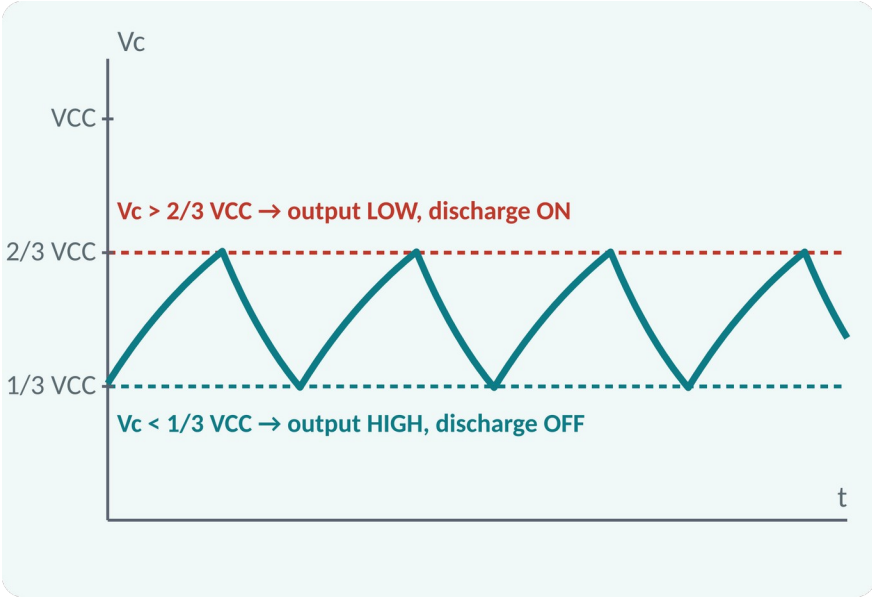


8-pin DIP; supply 4.5 V – 16 V

Supply 4.5–16 V; output current up to 200 mA

Operation: Set and Reset

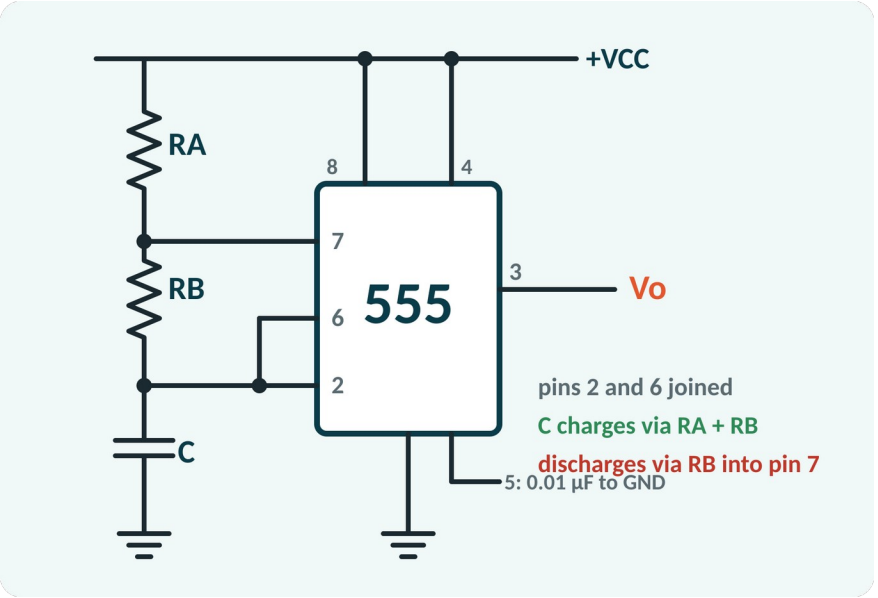
- Vc < 1/3 VCC: trigger comparator sets the flip-flop.
- Output HIGH; discharge transistor OFF; C charges.
- Vc > 2/3 VCC: threshold comparator resets it.
- Output LOW; discharge transistor ON; C discharges.
- Cycle repeats in astable mode.



Set at 1/3 VCC; reset at 2/3 VCC; Vc swings between the two

Astable Multivibrator Circuit

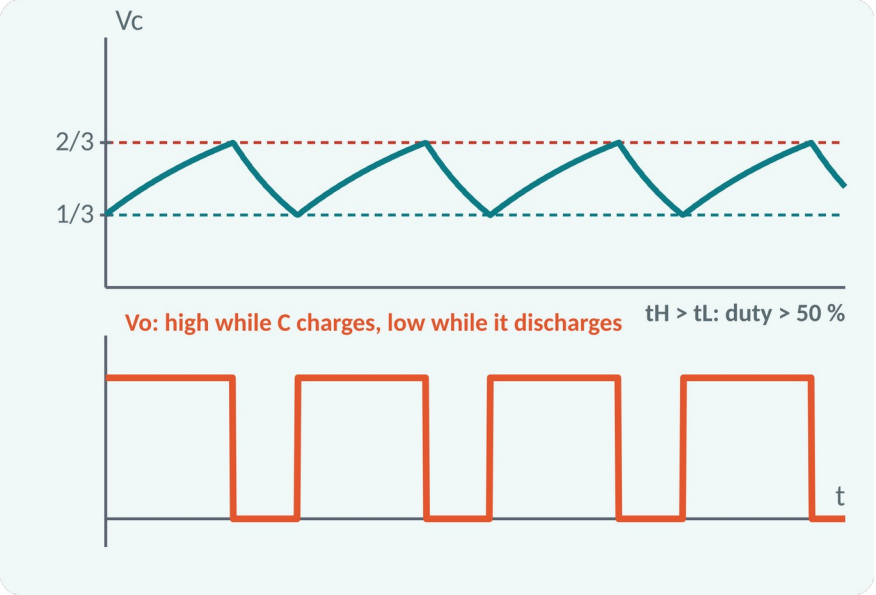
- External RA, RB and C set the timing.
- Pins 2 and 6 tied together to the capacitor.
- Pin 7 connects between RA and RB.
- No stable state: output toggles continuously.
- Also called a free-running oscillator.



Charging path: VCC → RA → RB → C; discharge path: C → RB → pin 7

Astable Waveforms

- C charges from 1/3 to 2/3 VCC through RA + RB.
- Output is HIGH during charging.
- C discharges from 2/3 to 1/3 VCC through RB.
- Output is LOW during discharging.
- Rectangular wave; duty cycle above 50 %.



Charging τ = (RA + RB)C; discharging τ = RB·C; interval = τ·ln 2 = 0.693τ

Timing Equations

- High time depends on RA + RB.
- Low time depends only on RB.
- Frequency from the total period.
- Duty cycle always above 50 %.
- Use RA ≪ RB for a near-50 % duty cycle.

$$t_H = 0.693 \cdot (R_A + R_B) \cdot C$$

$$t_L = 0.693 \cdot R_B \cdot C$$

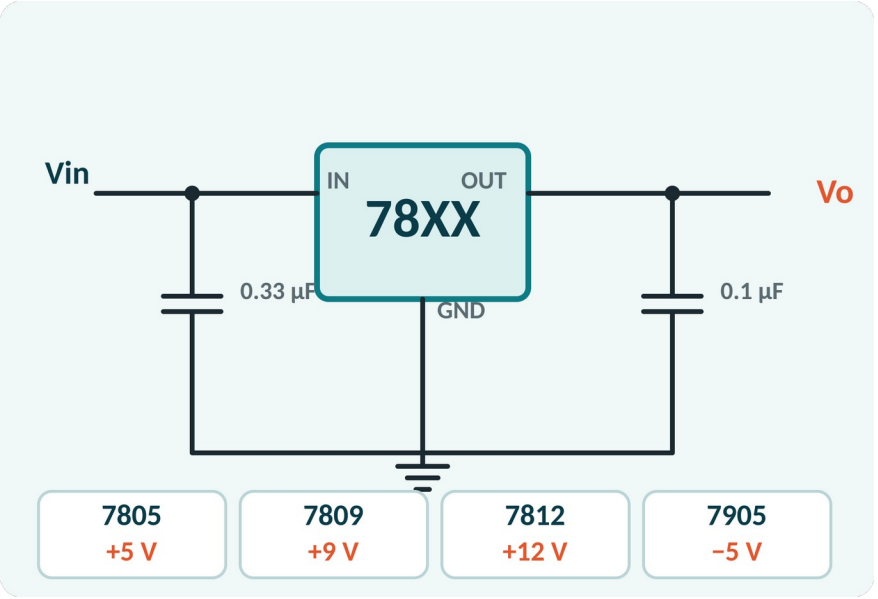
$$T = t_H + t_L = 0.693 \cdot (R_A + 2R_B) \cdot C$$

$$f = \frac{1.44}{(R_A + 2R_B) \cdot C}; \quad D = \frac{R_A + R_B}{R_A + 2R_B}$$

$$t_H = 0.693(R_A + R_B)C; \quad t_L = 0.693R_B \cdot C; \quad f = \frac{1.44}{(R_A + 2R_B)C}; \quad D = \frac{R_A + R_B}{R_A + 2R_B}$$

78XX Series Voltage Regulators

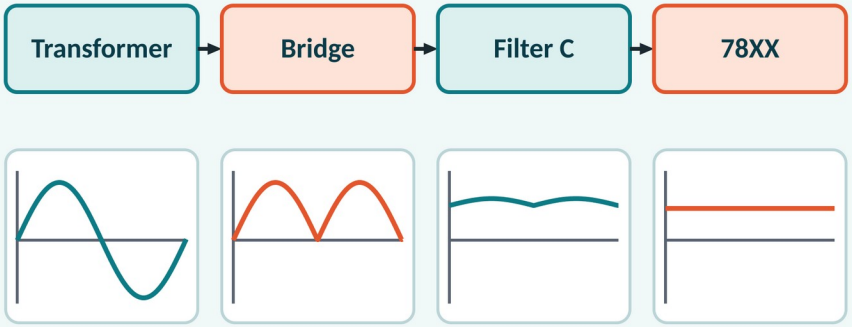
- Three-terminal fixed positive regulators.
- Last two digits give the output: 7805 = 5 V.
- Input must exceed output by about 2 V.
- Up to 1 A output with a heat sink.
- Built-in current limit and thermal shutdown.



Vo = fixed (5, 6, 8, 9, 12, 15, 18, 24 V); Vin(min) ≈ Vo + 2 V; 79XX: negative series

Regulator in a DC Power Supply

- Transformer lowers the mains voltage.
- Bridge rectifier gives pulsating DC.
- Capacitor filter smooths it.
- 78XX regulator gives fixed, ripple-free output.
- Regulator dissipates (Vin − Vo)·IL as heat.



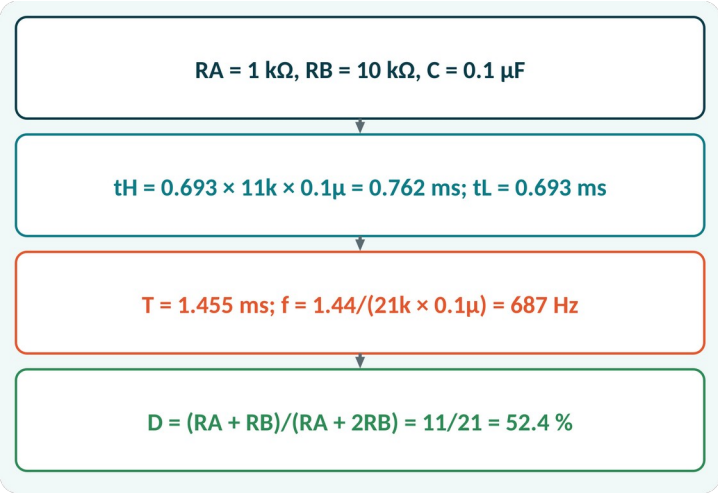
230 V AC → 12–15 V AC → pulsating DC → smoothed DC → fixed 5 V DC

Regulator keeps Vo constant for changes in Vin and load

Preg = (Vin − Vo)·IL; efficiency = Vo/Vin (linear regulator)

Solved Example 1: Astable Frequency

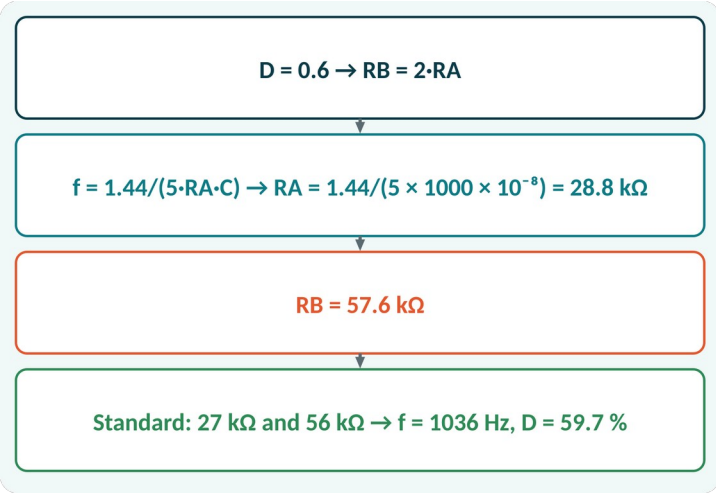
- A 555 astable has RA = 1 kΩ, RB = 10 kΩ, C = 0.1 µF. Find tH, tL, T, f and duty cycle.
- 1. tH = 0.693 × (1k + 10k) × 0.1 µF = 0.762 ms.
- 2. tL = 0.693 × 10k × 0.1 µF = 0.693 ms.
- 3. T = 1.455 ms; f = 1/T = 687 Hz.
- 4. D = tH/T = 0.762/1.455 = 52.4 %.



Answer: tH = 0.762 ms, tL = 0.693 ms, f = 687 Hz, duty = 52.4 %

Solved Example 2: Design for 1 kHz

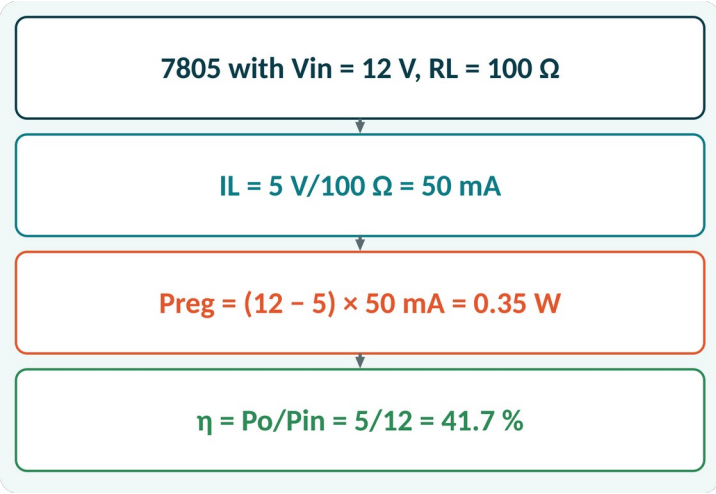
- Design an astable 555 for f = 1 kHz, duty cycle 60 %, C = 0.01 µF.
- 1. D = (RA + RB)/(RA + 2RB) = 0.6 → RB = 2RA.
- 2. f = 1.44/(RA + 2RB)C = 1.44/(5RA·C).
- 3. RA = 1.44/(5 × 1000 × 10^{-8}) = 28.8 kΩ; RB = 57.6 kΩ.
- 4. Standard values RA = 27 kΩ, RB = 56 kΩ: f = 1.44/(139k × 0.01 µF) = 1036 Hz; D = 59.7 %.



Answer: RA ≈ 28.8 kΩ, RB ≈ 57.6 kΩ (use 27 kΩ and 56 kΩ)

Solved Example 3: 7805 Regulator

- A 7805 is supplied from 12 V DC and feeds a 100 Ω load. Find IL, power dissipated in the regulator and efficiency. Is the input voltage adequate?
- 1. Vin(min) ≈ 5 + 2 = 7 V < 12 V: adequate.
- 2. IL = Vo/RL = 5/100 = 50 mA.
- 3. Preg = (Vin − Vo)·IL = 7 × 0.05 = 0.35 W.
- 4. η = Po/Pin = 5/12 = 41.7 %.



Answer: IL = 50 mA, Preg = 0.35 W, η ≈ 41.7 %

Common Mistakes and Exam Tips

Common mistakes

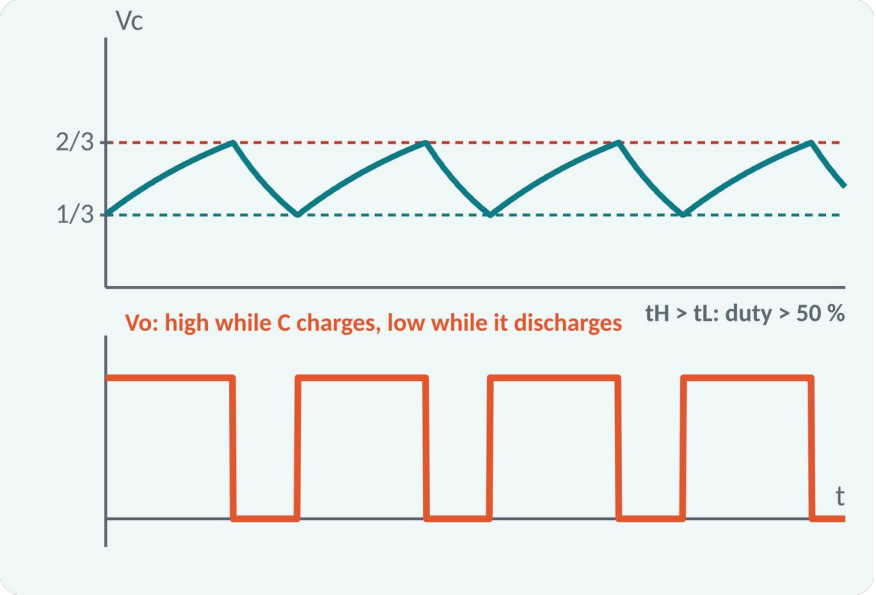
- Using tL with RA included.
- Mixing kΩ, µF and ms units.
- Forgetting to connect pin 4 to VCC.
- Expecting 50 % duty with RA = RB.
- Input too low for the 78XX (dropout).

Exam tips

- Joined pins 2 and 6 for astable operation.
- tH uses RA + RB; tL uses RB only.
- Vin ≥ Vo + 2 V for 78XX.
- Add input and output capacitors.

Summary

- 555: thresholds 2/3 VCC and 1/3 VCC; flip-flop and discharge transistor.
- Astable: C charges via RA + RB and discharges via RB.
- f = 1.44/[(RA + 2RB)C]; D = (RA + RB)/(RA + 2RB).
- 78XX: fixed output, three terminals, Vin ≥ Vo + 2 V.
- Linear regulator efficiency = Vo/Vin.



T = 0.693(RA + 2RB)C; Preg = (Vin − Vo)·IL

Quick Quiz

Q1. 555 comparator thresholds are:
(a) 1/2 and 1/4 VCC (b) 2/3 and 1/3 VCC (c) VCC and 0 (d) 3/4 and 1/4 VCC

Ans: (b)

Q2. In astable mode, C discharges through:
(a) RA (b) RA + RB (c) RB (d) none

Ans: (c)

Q3. RA = RB = 10 kΩ, C = 0.1 µF: f ≈
(a) 480 Hz (b) 1.44 kHz (c) 48 Hz (d) 4.8 kHz

Ans: (a)

Q4. 7812 output voltage is:
(a) 7 V (b) 12 V (c) 78 V (d) 1.2 V

Ans: (b)

Q5. Duty cycle of the standard astable circuit is always:
(a) below 50 % (b) exactly 50 % (c) above 50 % (d) 100 %

Ans: (c)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **David A. Bell, Operational Amplifier and Linear ICs**
   3rd Ed., Oxford University Press, 2011 (reference, Unit 5)

Next lecture: course revision